Linear Equations and Matrices

Dr. Juhi Kesarwani Dr. Ashish Kumar Kesarwany

Mathematics Division, School of Advanced Sciences and Languages
VIT Bhopal University, Bhopal-Indore Highway, Kothrikalan, Sehore, Madhya Pradesh, 466114, India

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Table of Contents

Notation:

$\mathbb{N}$: set of Natural numbers

$\mathbb{Z}$: set of integers

$\mathbb{Q}$: set of rational numbers

$\mathbb{R}$: set of real numbers

$\mathbb{C}$: set of complex numbers

$\mathbb{R}^2$: $\{(a, b) \ : \ a, b \in \mathbb{R}\}$ or $\{a\hat{i} + b\hat{j} \ : \ a, b \in \mathbb{R}\}$ or $\left\{ \begin{bmatrix} a \\ b \end{bmatrix} \ : \ a, b \in \mathbb{R} \right\}$

$\mathbb{R}^3$: $\{(a, b, c) : a, b, c \in \mathbb{R}\}$or $\{a\hat{i} + b\hat{j} + c\hat{k} : a, b, c \in \mathbb{R}\}$ or $\left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} : a, b, c \in \mathbb{R} \right\}$

$$\mathbb{R}^n : \left\{ \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \ : \ a_1, a_2, \ldots, a_n \in \mathbb{R} \right\}$$

$$\mathbb{C}^n : \left\{ \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \ : \ a_1, a_2, \ldots, a_n \in \mathbb{C} \right\}$$

The classical problem of linear algebra is to find all solutions (if any exist) to a **system of m linear equations in n unknowns** of the form

$$\left.\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + \cdots + a_{2n}x_n &= b_2 \\ \vdots \qquad \vdots \qquad \vdots \quad & \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m \end{aligned}\right\} \qquad (*)$$

Where a_{ij} and b_i are scalars belonging to $\mathbb{R}$ and x_j are variables which takes value in $\mathbb{R}$.

A system of linear equations of the above form is **homogeneous** if and only if $b_i = 0$ for all $1 \leq i \leq m$; otherwise it is **nonhomogeneous**.

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1. Does a given system of linear equations has a solution?
2. If it has a solution, is that solution unique?
3. If the solution is not unique, can we characterize the set of all solutions?
4. If there are solutions, how do we compute them efficiently?

We can write the system of linear equations $(*)$ in the following form

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

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- The matrix $A = [a_{ij}]$ is called the **coefficient matrix** of the system.
- The matrix $A^* = [A \mid \mathbf{b}]$ is called the **augmented matrix** of the system.

* A vector $\mathbf{c} = \begin{bmatrix} c_1 & \cdots & c_n \end{bmatrix}^T \in \mathbb{R}^n$ is called the solution of the system if $A\mathbf{c} = \mathbf{b}$.

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- Set of all vectors satisfying $AX = \mathbf{b}$ is called the **solution set of the system**.

- If a system $AX = \mathbf{b}$ has a solution then it is called **consistent** otherwise it is called **inconsistent**. A homogeneous system is always consistent because $\begin{bmatrix} 0 & \cdots & 0 \end{bmatrix}^T \in \mathbb{R}^n$ is always a solution of $AX = \mathbf{0}$.

Elementary row operations

There are three types of elementary row operations

Type-1: Swapping/Interchanging any two rows of a matrix, i.e.,

$$R_i \longleftrightarrow R_j$$

Type-2: Multiplying a row by a non-zero constant,, i.e.,

$$R_i \to \alpha R_i \qquad \alpha \neq 0$$

Type-3: Replacing a row with the sum of itself and a multiple of another row. , i.e.,

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Leading entry or Pivot element

The first non-zero element of a row of a matrix is called leading entry or leading element or pivot element or pivot entry of that row.

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- **Rank of a Matrix:** The rank of a matrix is defined to be the number of non-zero rows in its row echelon form.

- **Remark:**

  1. REF of a matrix is not unique but # of non-zero rows and # zeros rows in any two REFs is same.
  2. We can apply any two elementary row operations simultaneously only if they are not effecting each other.

Example

Find the rank of the matrix

$$A = \begin{bmatrix} 2 & 3 & 0 & 1 \\ 1 & 0 & 1 & 2 \\ -1 & 1 & 1 & -2 \\ 1 & 5 & 3 & -1 \end{bmatrix}.$$

Solution

Reduce the given matrix A to a row echelon form by applying elementary row operations. Given:

$$A = \begin{bmatrix} 2 & 3 & 0 & 1 \\ 1 & 0 & 1 & 2 \\ -1 & 1 & 1 & -2 \\ 1 & 5 & 3 & -1 \end{bmatrix}$$

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$$R_4 \to R_4 - R_3 \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & -8 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

which is in row echelon form and therefore

$$\text{rank}(A) = \# \text{ non-zero rows in its row echelon form } = 3.$$

Exercise 1:

Compute the rank of the following matrices

$$A_2 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \qquad A_3 = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \qquad A_4 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{pmatrix}$$

Can you generalize this and guess the rank of A_n for $n \geq 2$? Justify your argument.

Exercise 2:

Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$. Ans: $\text{rank}(A) = 3.$

Exercise 3:

Find the rank of matrix : $A = \begin{bmatrix} 1 & 2 & 3 & -4 \\ -2 & 3 & 7 & 1 \\ 1 & 9 & 16 & -11 \end{bmatrix}$. Ans: $\text{rank}(A) = 2.$

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(3). *The system $Ax = \mathbf{b}$ has infinitely many solutions iff* $\text{rank}(A) = \text{rank}(A^*) < n$.

Example

Consider the following system of linear equations

$$\begin{aligned} 2y + z &= -8 \\ x - 2y - 3z &= 0 \\ -x + y + 2z &= 3 \end{aligned}$$

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$$A^* = \left[\begin{array}{rrr|r} 0 & 2 & 1 & -8 \\ 1 & -2 & -3 & 0 \\ -1 & 1 & 2 & 3 \end{array}\right]$$

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Here $\text{rank}(A) = \text{rank}(A^*) = \#$ varables. Hence system is consistent and it has unique solution.

Example

Consider the following system of linear equations

$$\begin{aligned} x - y + 2z &= -3 \\ 4x + 4y - 2z &= 1 \\ -2x + 2y - 4z &= 6 \end{aligned}$$

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The augmented matrix of given system is

$$C^* = \left[\begin{array}{rrr|r} 1 & -1 & 2 & -3 \\ 4 & 4 & -2 & 1 \\ -2 & 2 & -4 & 6 \end{array}\right]$$

Example

Consider the following system of linear equations

$$
\begin{aligned}
x - y + 2z &= -3 \\
4x + 4y - 2z &= 1 \\
-2x + 2y - 4z &= 6
\end{aligned}
$$

The augmented matrix of given system is

$$
C^* = \left[\begin{array}{rrr|r} 1 & -1 & 2 & -3 \\ 4 & 4 & -2 & 1 \\ -2 & 2 & -4 & 6 \end{array}\right] \begin{array}{l} R_2 \to R_2 - 4R_1 \\ R_3 \to R_3 + 2R_1 \end{array}
$$

Example

Consider the following system of linear equations

$$\begin{aligned} x - y + 2z &= -3 \\ 4x + 4y - 2z &= 1 \\ -2x + 2y - 4z &= 6 \end{aligned}$$

The augmented matrix of given system is

$$C^* = \left[\begin{array}{rrr|r} 1 & -1 & 2 & -3 \\ 4 & 4 & -2 & 1 \\ -2 & 2 & -4 & 6 \end{array}\right] \begin{array}{l} R_2 \to R_2 - 4R_1 \\ R_3 \to R_3 + 2R_1 \end{array} \left[\begin{array}{rrr|r} 1 & -1 & 2 & -3 \\ 0 & 8 & -10 & 13 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

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Here $\text{rank}(C) = \text{rank}(C^*) < \#$ variables, therefore system has infinitely many solutions.

Example

Consider the following system of linear equations

$$
\begin{aligned}
x_1 + x_2 &= 1\\
2x_1 + 2x_2 &= 3\\
3x_1 + 3x_2 &= 3.
\end{aligned}
$$

Example

Consider the following system of linear equations

$$\begin{aligned} x_1 + x_2 &= 1 \\ 2x_1 + 2x_2 &= 3 \\ 3x_1 + 3x_2 &= 3. \end{aligned}$$

Then the augmented matrix is

$$A^* = \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{array}\right].$$

Example

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We reduce this by elementary row operations as follows.

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{array}\right]$$

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$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{array}\right] \xrightarrow[]{\substack{R_2-2R_1 \\ R_3-3R_1}}$$

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Example

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$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{array}\right] \xrightarrow[]{\substack{R_2-2R_1 \\ R_3-3R_1}} \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array}\right].$$

The last matrix is in echelon form and it has two nonzero rows. Thus, $\text{rank}(A) = 1$ and $\text{rank}(A^*) = 2$. Since $\text{rank}(A) \neq \text{rank}(A^*)$ and hence system is inconsistent.

Example

For what values of $\lambda \in \mathbb{R}$, the following system of equations has

(i) no solution,

(ii) a unique solution, and

(iii) infinitely many solutions?

$$\begin{aligned} x + y + \lambda z &= 1, \\ x + \lambda y + z &= 1, \\ \lambda x + y + z &= -2. \end{aligned}$$

Solution

The Augmented matrix

$$A^* = \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 1 & \lambda & 1 & 1 \\ \lambda & 1 & 1 & -2 \end{array}\right]$$

Solution

The Augmented matrix

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Solution

The Augmented matrix

$$A^* = \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 1 & \lambda & 1 & 1 \\ \lambda & 1 & 1 & -2 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - \lambda R_1 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 0 & (\lambda - 1) & (1 - \lambda) & 0 \\ 0 & (1 - \lambda) & (1 - \lambda^2) & (-2 - \lambda) \end{array}\right]$$

Solution

The Augmented matrix

$$A^* = \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 1 & \lambda & 1 & 1 \\ \lambda & 1 & 1 & -2 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - \lambda R_1 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 0 & (\lambda - 1) & (1 - \lambda) & 0 \\ 0 & (1 - \lambda) & (1 - \lambda^2) & (-2 - \lambda) \end{array}\right]$$

$$R_3 \to R_3 + R_2$$

Solution

The Augmented matrix

$$A^* = \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 1 & \lambda & 1 & 1 \\ \lambda & 1 & 1 & -2 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - \lambda R_1 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 0 & (\lambda - 1) & (1 - \lambda) & 0 \\ 0 & (1 - \lambda) & (1 - \lambda^2) & (-2 - \lambda) \end{array}\right]$$

$$R_3 \to R_3 + R_2 \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 0 & (\lambda - 1) & (1 - \lambda) & 0 \\ 0 & 0 & (1 - \lambda)(2 + \lambda) & (-2 - \lambda) \end{array}\right],$$

Solution

The Augmented matrix

$$A^* = \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 1 & \lambda & 1 & 1 \\ \lambda & 1 & 1 & -2 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - \lambda R_1 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 0 & (\lambda - 1) & (1-\lambda) & 0 \\ 0 & (1-\lambda) & (1-\lambda^2) & (-2-\lambda) \end{array}\right]$$

$$R_3 \to R_3 + R_2 \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 0 & (\lambda - 1) & (1-\lambda) & 0 \\ 0 & 0 & (1-\lambda)(2+\lambda) & (-2-\lambda) \end{array}\right],$$

which is in row echelon form.

Solution

The Augmented matrix

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which is in row echelon form.

- **No Solution:** In this case, $\text{rank}(A) \neq \text{rank}(A^*)$. Thus if $(1-\lambda)(2+\lambda) = 0$ and $(-2-\lambda) \neq 0$. That is, if $\lambda = 1$, then the given system has no solution.

Solution

The Augmented matrix

$$A^* = \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 1 & \lambda & 1 & 1 \\ \lambda & 1 & 1 & -2 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - \lambda R_1 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 0 & (\lambda - 1) & (1 - \lambda) & 0 \\ 0 & (1 - \lambda) & (1 - \lambda^2) & (-2 - \lambda) \end{array}\right]$$

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- **Unique Solution:** In this case, $\text{rank}(A) = \text{rank}(A^*) = 3$. Thus $(1 - \lambda)(2 + \lambda) \neq 0$. That is, if $\lambda \neq 1, -2$, then given system has unique solution.

Solution

The Augmented matrix

$$A^* = \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 1 & \lambda & 1 & 1 \\ \lambda & 1 & 1 & -2 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - \lambda R_1 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & \lambda & 1 \\ 0 & (\lambda-1) & (1-\lambda) & 0 \\ 0 & (1-\lambda) & (1-\lambda^2) & (-2-\lambda) \end{array}\right]$$

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- **Infinitely many solutions:** In this case, $\text{rank}(A) = \text{rank}(A^*) < 3$. Thus if $(1-\lambda)(2+\lambda) = 0$ and $(-2-\lambda) = 0$. That is, if $\lambda = -2$, then $\text{rank}(A) = \text{rank}(A^*) = 2$ and hence given system has infinitely many solutions.

Example

Consider the following system of linear equations

$$\begin{aligned} x - 2y + z &= 3 \\ x + 2y + az &= 6 \\ x + ay + 3z &= b. \end{aligned}$$

For which values of a does the system have unique solution and for which pair (a, b) does the system has more than one solution.

Solution

The augmented matrix

$$A^* = \begin{bmatrix} 1 & -2 & 1 & | & 3 \\ 1 & 2 & a & | & 6 \\ 1 & a & 3 & | & b \end{bmatrix} \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix} \begin{bmatrix} 1 & -2 & 1 & | & 3 \\ 0 & 4 & a-1 & | & 3 \\ 0 & a+2 & 2 & | & b-3 \end{bmatrix}$$

$$R_3 \to R_3 - \left(\tfrac{a+2}{4} R_2\right) \begin{bmatrix} 1 & -2 & 1 & | & 3 \\ 0 & 4 & a-1 & | & 3 \\ 0 & 0 & 2 - \frac{(a-1)(a+2)}{4} & | & (b-3) - \frac{3}{4}(a+2) \end{bmatrix}.$$

Solution

The augmented matrix

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- **Unique Solution:** In this case, $\text{rank}(A) = \text{rank}(A^*) = 3$. Thus $2 - \frac{(a-1)(a+2)}{4} \neq 0$. That is, if $a \neq \dfrac{-1 \pm \sqrt{41}}{2}$, then given system has unique solution.

Solution

The augmented matrix

$$A^* = \begin{bmatrix} 1 & -2 & 1 & | & 3 \\ 1 & 2 & a & | & 6 \\ 1 & a & 3 & | & b \end{bmatrix} \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix} \begin{bmatrix} 1 & -2 & 1 & | & 3 \\ 0 & 4 & a-1 & | & 3 \\ 0 & a+2 & 2 & | & b-3 \end{bmatrix}$$

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- **Unique Solution:** In this case, $\text{rank}(A) = \text{rank}(A^*) = 3$. Thus $2 - \frac{(a-1)(a+2)}{4} \neq 0$. That is, if $a \neq \dfrac{-1 \pm \sqrt{41}}{2}$, then given system has unique solution.

- **Infinitely many solutions:** In this case, $\text{rank}(A) = \text{rank}(A^*) < 3$. Thus if $2 - \frac{(a-1)(a+2)}{4} = 0$ and $(b-3) - \frac{3}{4}(a+2) = 0$. That is, if $a = \dfrac{-1 \pm \sqrt{41}}{2}$ and $b = \dfrac{33 \pm 3\sqrt{41}}{8}$, then given system has infinitely many solutions.

Exercise 4:

For which value of $k \in \mathbb{R}$, the system of equations

$$\begin{aligned} x + y + z &= 1 \\ 2x + 3y - z &= 5 \\ x + 2y - kz &= 4 \end{aligned}$$

has infinite number of solutions. **Ans:** $k = 2$.

Exercise 5:

Suppose that the following matrix A is the augmented matrix for a system of linear equations.

$$A = \left[\begin{array}{rrr|r} 1 & 2 & 3 & 4 \\ 2 & -1 & -2 & a^2 \\ -1 & -7 & -11 & a \end{array}\right],$$

where a is a real number. Determine all the values of a so that the corresponding system is consistent. **Ans:** $a = -3, 4$

Exercise 6:
For what values of $\lambda \in \mathbb{R}$, the following system of equations has (i) no solution, (ii) a unique solution, and (iii) infinitely many solutions?

$$x - 2y + 3z = 2, x + y + z = \lambda, 2x - y + 4z = \lambda^2.$$

Ans: for no solution $\lambda \neq 2, -1$, no unique solution and for infinitely many solutions $\lambda = 2, -1$.

Solution

The Augmented matrix

$$A^* = \begin{bmatrix} 1 & -2 & 3 & | & 2 \\ 1 & 1 & 1 & | & \lambda \\ 2 & -1 & 4 & | & \lambda^2 \end{bmatrix} \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - 2R_1 \end{matrix} \begin{bmatrix} 1 & -2 & 3 & | & 2 \\ 0 & 3 & -2 & | & (\lambda - 2) \\ 0 & 3 & -2 & | & (\lambda^2 - 4) \end{bmatrix}$$

$$R_3 \to R_3 - R_2 \begin{bmatrix} 1 & -2 & 3 & | & 2 \\ 0 & 3 & -2 & | & (\lambda - 2) \\ 0 & 0 & 0 & | & (\lambda - 2)(\lambda + 1) \end{bmatrix},$$ which is in row echelon form.

- **No Solution:** In this case, the rank of the coefficient matrix $\neq$ the rank of the augmented matrix. Thus if $(\lambda - 2)(\lambda + 1) \neq 0$. That is, if $\lambda \neq 2, -1$ then the given system has no solution.
- **Unique Solution:** In this case, the rank of the coefficient matrix $=$ the rank of the augmented matrix $= 3$. Notice that $rank(A) = 2$ for any value of λ. Thus the then given system has no unique solution.
- **Infinitely many solutions:** In this case, the rank of the coefficient matrix $=$ the rank of the augmented matrix < 3. Thus if $(\lambda - 2)(\lambda + 1) = 0$. That is if $\lambda = 2, -1$ then $rank(A) = rank(A^*) = 2$ and hence given system has infinitely many solutions.

Exercise 7:

In the following cases find out the conditions on b_i's so that the system is consistent / inconsistent.

1. $A = \begin{bmatrix} 1 & -3 & -4 \\ -3 & 2 & 6 \\ 5 & -1 & -8 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

   Ans: The system is consistent iff $b_3 + 2b_2 + b_1 = 0$.

2. $A = \begin{bmatrix} 1 & 2 & 3 & 5 \\ 2 & 4 & 8 & 12 \\ 3 & 6 & 7 & 13 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

   Ans: The system is consistent iff $b_3 + b_2 - 5b_1 = 0$.

Exercise 8:

Consider the following augmented matrix in which $*$ denotes an arbitrary number and $\blacksquare$ denotes a nonzero number. Determine whether the given augmented matrix is consistent. If consistent, is the solution unique?

1. $\left[\begin{array}{ccccc|c} \blacksquare & * & * & * & * & * \\ 0 & \blacksquare & * & * & 0 & * \\ 0 & 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * \end{array}\right]$

2. $\left[\begin{array}{ccc|c} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & \blacksquare & * \end{array}\right]$

3. $\left[\begin{array}{ccccc|c} \blacksquare & * & * & * & * & * \\ 0 & \blacksquare & 0 & * & 0 & * \\ 0 & 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * \end{array}\right]$

4. $\left[\begin{array}{ccccc|c} \blacksquare & * & * & * & * & * \\ 0 & \blacksquare & * & * & 0 & * \\ 0 & 0 & 0 & 0 & \blacksquare & 0 \\ 0 & 0 & 0 & 0 & * & \blacksquare \end{array}\right]$

Ans:

1. **The solution exists but is not unique.**
2. **A solution exists and is unique.**
3. **There are infinitely many solutions.**
4. **There might be a solution. If so, there are infinitely many.**

Geometric Interpretations

1. The two lines are parallel (and not the same), so there are no solutions.
2. The two lines intersect in a point, so there is one solution.
3. The two lines are the same, so there are an infinite number of solutions.

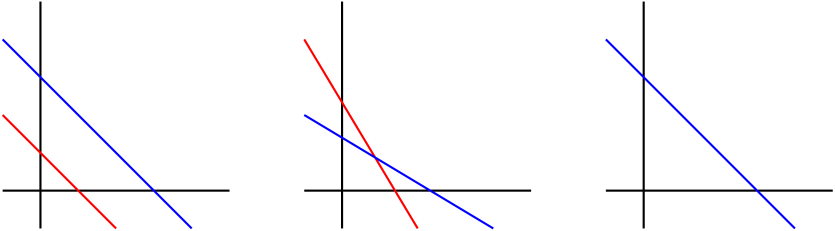

Figure 1: No solution, unique solution, and infinitely many solutions.

In three variable case following are the possibilities:

There is unique solution. In order for three equations with three variables to have unique solution, the planes must intersect in a single point.

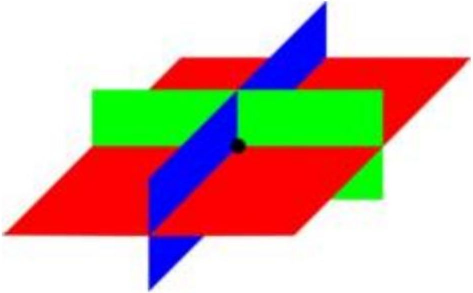

Figure 2: Unique solution

There is no solution. The three planes do not have any points in common. There are following three possibilities:

1. The planes are different, but all parallel.
2. The planes are different and none are parallel. but the lines of intersection of each pair are parallel.
3. Two planes are parallel, the third crosses them.

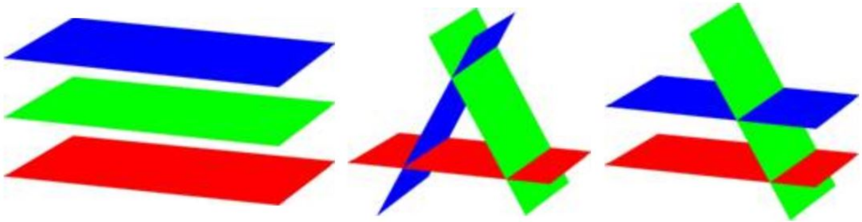

Figure 3: No solution

There are an infinite number of solutions. There are following two possibilities:

1. Three planes intersect in a line.
2. All three planes are the same.

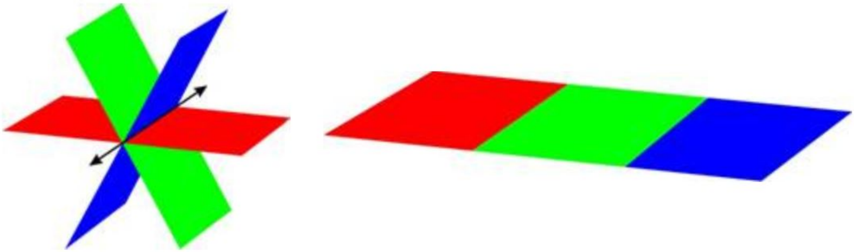

Figure 4: Infinitely many solutions solution

Methods for solving System of Linear Equations

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Definition (Gaussian Elimination)

Consider a system of linear equations $AX = \mathbf{b}$. The method of solving this system by **Gaussian elimination** with back-substitution equation is described as follows:

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1. Write the augmented matrix of the system.
2. Use the elementary row operations to reduce the augmented matrix to a matrix in row-echelon form.
3. Write the linear system corresponding to the row-echelon matrix and solve by back-substitution.

Example

We will use the method of Gaussian elimination with back-substitution to solve following system of linear equations using sing analogous steps

$$\begin{aligned} x_1 + 3x_4 &= 4 \\ 2x_2 - x_3 - x_4 &= 0 \\ 3x_2 - 2x_4 &= 1 \\ 2x_1 - x_2 + 4x_3 &= 5 \end{aligned}$$

The augmented matrix is:

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 4 \\ 0 & 2 & -1 & -1 & 0 \\ 0 & 3 & 0 & -2 & 1 \\ 2 & -1 & 4 & 0 & 5 \end{array}\right]$$

Converting the above matrix in row echelon form, we have

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 4 \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 0 & 1 & 1 \end{array}\right]$$

From here we see that $\text{rank}(A) = \text{rank}(A^*) = 4$ and hence given system is consistent and has a unique solution.

Now the system of linear equations corresponding this row-echelon matrix is

$$\begin{aligned} x_1 + 3x_4 &= 4 \\ x_2 - \frac{1}{2}x_3 - \frac{1}{2}x_4 &= 0 \\ x_3 - \frac{1}{3}x_4 &= \frac{2}{3} \\ x_4 &= 1 \end{aligned}$$

By back-substitution:

$$x_1 = x_2 = x_3 = x_4 = 1$$

Exercise 9:
Solve the following system of linear equations by transforming its augmented matrix to row echelon form (Gaussian elimination).

1.

$$\begin{aligned} x_1 - x_3 - 3x_5 &= 1 \\ 3x_1 + x_2 - x_3 + x_4 - 9x_5 &= 3 \\ x_1 - x_3 + x_4 - 2x_5 &= 1. \end{aligned}$$

Ans: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} s + \begin{bmatrix} 3 \\ 1 \\ 0 \\ -1 \\ 1 \end{bmatrix} t + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, for any $s, t \in \mathbb{R}$

2.

$$\begin{aligned} x_1 + x_2 - x_5 &= 1 \\ x_2 + 2x_3 + x_4 + 3x_5 &= 1 \\ x_1 - x_3 + x_4 + x_5 &= 0 \end{aligned}$$

Ans: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 3 \\ -2 \end{bmatrix} s + \begin{bmatrix} -6 \\ 7 \\ -5 \end{bmatrix} t + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, for any $s, t \in \mathbb{R}$

3.

$$\begin{aligned} 6x + 8y + 6z + 3w &= -3 \\ 6x - 8y + 6z - 3w &= 3 \\ 8y \qquad - 6w &= 6 \end{aligned}$$

Ans: $(x, y, z, w) = (-t, 0, t, -1)$, for any $t \in \mathbb{R}$

Gauss-Jordan method

❋ **Row reduced Echelon form (RREF) or Gauss-Jordan form:** A matrix is said to be in row reduced echelon form (rref) or Gauss-Jordan form if

1. It is in row-echelon form.
2. All the leading entries are 1.
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- **Rank of a matrix:** The rank of a matrix is defined to be the number of non-zero rows in its RREF.

- **Remark:** RREF of a matrix is unique.

Example

Which of the following matrices are in row echelon form? Which are in reduced row echelon form?

(1) $\begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

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(1) $\begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ Neither REF nor RREF

(2) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

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Which of the following matrices are in row echelon form? Which are in reduced row echelon form?

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Which of the following matrices are in row echelon form? Which are in reduced row echelon form?

(1) $\begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ Neither REF nor RREF

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(3) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ REF

(4) $\begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 \end{bmatrix}$

(4) $\begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 \end{bmatrix}$ Neither REF nor RREF

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Neither REF nor RREF

(5) $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

(4) $\begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 \end{bmatrix}$ Neither REF nor RREF

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(6) $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

(4) $\begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 \end{bmatrix}$ Neither REF nor RREF

(5) $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ REF

(6) $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ Neither REF nor RREF

Exercise 10:

List all possible row reduced Echelon form of each of a 2×2 and 3×3 matrix.

Hint: For 2×2 matrix: 4, and for 3×3 matrix: 8.

Definition (Gauss-Jordan elimination)

The method of Gaussian elimination with back substitution to solve system of linear equations can be refined by first further reducing the augmented matrix to a Gauss-Jordan form and work with the system corresponding to it. This method is called Gauss-Jordan elimination method of solving linear systems.

Example

Consider the following system of linear equations

$$\begin{aligned} x_1 + x_3 &= 2 \\ 2x_1 + 6x_2 - 2x_3 &= 3 \\ 4x_1 + 8x_2 - 5x_3 &= 4 \end{aligned}$$

The augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 2 & 6 & -2 & 3 \\ 4 & 8 & -5 & 4 \end{array}\right]$$

The RREF of this matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 14/11 \\ 0 & 1 & 0 & 7/22 \\ 0 & 0 & 1 & 8/11 \end{array}\right]$$

From here we see that $\text{rank}(A) = \text{rank}(A^*) = 3$ and hence this system is consistent and has unique solution. The above RREF corresponds to the augmented matrix of the following system.

$$x_1 = \frac{14}{11}$$
$$x_2 = \frac{7}{22}$$
$$x_3 = \frac{8}{11}$$

This gives the unique solution of above system of linear equations.

Exercise 11:

Solve the following system of linear equations using Gauss-Jordan method

1. $\begin{bmatrix} 1 & 1 & -3 \\ 2 & 1 & 4 \\ 5 & 2 & 16 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 4 \end{bmatrix}$ Ans: $x = -10, y = 19, z = 1$

2. $$\begin{aligned} x + 3y + 5z &= 6 \\ 6x - 8y + 4z &= -3 \\ 3x + 11y + 13z &= 17 \end{aligned}$$

   Ans: $x = -\frac{1}{2}, y = \frac{1}{2}, z = 1$

3. $$\begin{aligned} x + 5y + 7z &= 41 \\ 5x - 4y + 6z &= 2 \\ 7x + 9y - 3z &= 1 \end{aligned}$$

   Ans: $x = -2, y = 3, z = 4$

4.

$$\begin{aligned} x + 3y + 2z &= 14 \\ 2x + y + z &= 7 \\ 3x + 2y - z &= 7 \end{aligned}$$

Ans: $x = 1, y = 3, z = 2$

5.

$$\begin{aligned} x + y + 2z &= 2 \\ 2x - y + z &= -2 \\ 3x + y + 4z &= 2 \end{aligned}$$

Ans: $x = -t, y = 2 - t, z = t$, for any $t \in \mathbb{R}$

6.

$$\begin{aligned} x + 2y + z &= 1 \\ x + y + 3z &= 2 \\ 3x + 5y + 5z &= 4 \end{aligned}$$

Ans: $x = 3 - 5t, y = 2t - 1, z = t$, for any $t \in \mathbb{R}$

7.

$$\begin{aligned} 3x - 2y - 18z &= 6 \\ 2x + y - 5z &= 25 \end{aligned}$$

Ans: $x = 8 + 4t, y = 9 - 3t, z = t,$ for any $t \in \mathbb{R}$

8.

$$\begin{aligned} x + 5y + 2z &= 9 \\ 2x - y + 2z &= 4 \end{aligned}$$

Ans: $x = 12t + 7, y = 2t + 2, z = -11t - 4,$ for any $t \in \mathbb{R}$

9.

$$\begin{aligned} x + y + z &= 0 \\ 2x + 4z + w &= -1 \\ 3x + 2y + 4z + w &= 0 \end{aligned}$$

Ans: $\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \end{bmatrix} t,$ for any $t \in \mathbb{R}$

10

$$
\begin{aligned}
x + 3y + z &= 9\\
x + y - z &= 1\\
3x + 11y + 5z &= 35
\end{aligned}
$$

Ans: $(x, y, z) = (2t - 3, 4 - t, t)$, for any $t \in \mathbb{R}$

Solution set for homogeneous system $AX = \mathbf{0}$

The following theorem gives the solution set of homogeneous system of linear equations:

Theorem

Let A be a $m \times n$ coefficient matrix of homogeneous system $AX = \mathbf{0}$ with $\operatorname{rank}(A) = r$. Then there are $n - r$ linearly independent solutions of system $AX = \mathbf{0}$.
If $v_1, \ldots, v_{n-r}$ are $n - r$ linearly independent solutions of system. Then

$$S = \left\{ \sum_{i=1}^{n-r} \alpha_i v_i \mid \alpha_i \in \mathbb{R} \right\}$$

is the solution set of homogeneous system $AX = \mathbf{0}$.

Example

Consider the following homogeneous system of linear equations

$$\begin{aligned} x_1 - x_2 + 2x_3 &= 0 \\ 2x_1 + x_2 + x_3 &= 0 \\ x_1 + x_2 &= 0 \end{aligned}$$

The augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{array}\right]$$

The REF of this matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Hence, from here we see that $\text{rank}(A) = 2$ and hence using above theorem we see that this system has $3 - 2 = 1$ linearly independent solution. The above REF corresponds to the augmented matrix of the following system.

$$x_1 + x_3 = 0 \implies x_1 = -x_3$$
$$x_2 - x_3 = 0 \implies x_2 = x_3$$

So, we can describe all solution using x_3. Let $x_3 = t$, so solution set for this system is

$$\left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mid x_1 = -t, x_2 = t, x_3 = t \right\} = \left\{ \begin{bmatrix} -t \\ t \\ t \end{bmatrix} \mid t \in \mathbb{R} \right\} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} t \mid t \in \mathbb{R} \right\}$$

Solution set for nonhomogeneous system $AX = \mathbf{b}$

The following theorem gives the solution set of nonhomogeneous system $AX = \mathbf{b}$

Theorem

Let $AX = \mathbf{b}$ *be a nonhomogeneous system of linear equations. If* $u \in \mathbb{R}^n$ *is a solution of this system, and if* S *is solution set of associated homogeneous system,then the set*

$$u + S = \{u + v \mid v \in S\}$$

is solution set for nonhomogeneous part of this system.

Solution set for nonhomogeneous system $AX = \mathbf{b}$

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is solution set for nonhomogeneous part of this system.

Or

In other words, if $\operatorname{rank}(A) = r$ *and* $v_1, \ldots, v_{n-r}$ *be linearly independent solution of homogeneous part of this system. Then*

$$T = \left\{ u + \sum_{i=1}^{n-r} \alpha_i v_i \mid \alpha_i \in \mathbb{R} \right\}$$

is the solution set of nonhomogeneous system $AX = \mathbf{b}$*.*

Example

Consider the following nonhomoeneous system of linear equations

$$\begin{aligned} 2x_1 - x_2 + 3x_3 &= 5 \\ 3x_1 + 2x_2 - 2x_3 &= 1 \\ 7x_1 + 4x_3 &= 11 \end{aligned}$$

The augmented matrix is :

$$\left[\begin{array}{ccc|c} 2 & -1 & 3 & 5 \\ 3 & 2 & -2 & 1 \\ 7 & 0 & 4 & 11 \end{array}\right]$$

The RREF of this matrix is :

$$\left[\begin{array}{ccc|c} 1 & 0 & 4/7 & 11/7 \\ 0 & 1 & -13/7 & -13/7 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

From here, we see that $\text{rank}(A) = \text{rank}(A^*) = 2 < 3$, hence this system is consistent and has infinitely many solutions. The homogeneous part of this system has $3-2 = 1$ linearly independent solution namely $\begin{bmatrix} -4/7 \\ 13/7 \\ 1 \end{bmatrix}$. The above RREF corresponds to augmented matrix of following system

$$x_1 + \frac{4}{7}x_3 = \frac{11}{7}$$
$$x_2 - \frac{13}{7}x_3 = -\frac{13}{7}$$

If we put $x_3 = 0$ then we see that $\begin{bmatrix} 11/7 \\ -13/7 \\ 0 \end{bmatrix}$ is a particular Solution of this system.

Hence Solution set for this system is

$$S = \left\{ \begin{bmatrix} 11/7 \\ -13/7 \\ 0 \end{bmatrix} + t \begin{bmatrix} -4/7 \\ 13/7 \\ 1 \end{bmatrix} \mid t \in \mathbb{R} \right\}$$

Gauss-Jordan Method for finding the inverse of a matrix

Let A be an invertible matrix of size $n \times n$ whose inverse we want to find. The Gauss-Jordan elimination method works as follows:

1. We will start with two matrices. We want to invert the matrix A alongside another matrix that we will set at the start of the process to the identity matrix.

$$[A|I_n]$$

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3. Once the process is finished, the second matrix will end up being A^{-1} our inverted matrix.

$$\left[I_n|A^{-1}\right]$$

Geometrical Representation

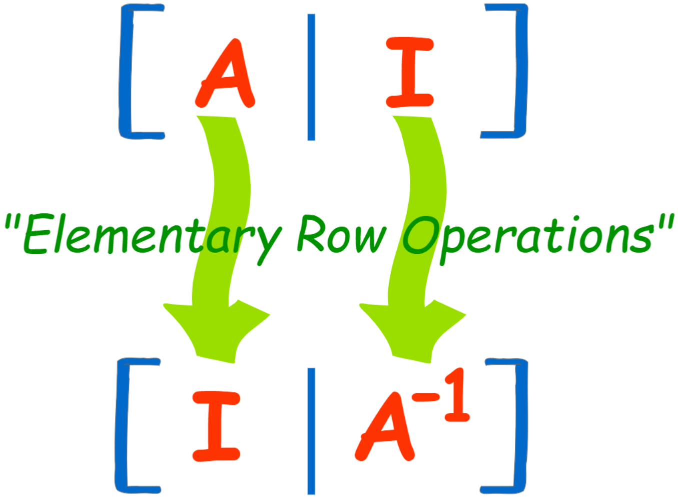

Figure 5: Gauss-Jordan method for finding inverse of a matrix

Example

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ by using Gauss-Jordan method.

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Solution

$[A|I_n]$

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Solution

$$[A|I_n] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array}\right]$$

Example

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ by using Gauss-Jordan method.

Solution

$$[A|I_n] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{array}$$

Example

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ by using Gauss-Jordan method.

Solution

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Example

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ by using Gauss-Jordan method.

Solution

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Example

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ by using Gauss-Jordan method.

Solution

$$[A|I_n] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array}\right] R_3 \to R_3 - R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 2 & 0 & -1 & 1 \end{array}\right]$$

Example

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ by using Gauss-Jordan method.

Solution

$$[A|I_n] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array}\right] R_3 \to R_3 - R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 2 & 0 & -1 & 1 \end{array}\right] \begin{array}{c} R_1 \to R_1 - R_2 \\ R_3 \to \frac{R_3}{2} \end{array}$$

Example

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ by using Gauss-Jordan method.

Solution

$$[A|I_n] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array}\right] R_3 \to R_3 - R_2$$

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Example

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ by using Gauss-Jordan method.

Solution

$$[A|I_n] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array}\right] R_3 \to R_3 - R_2$$

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Example

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ by using Gauss-Jordan method.

Solution

$$[A|I_n] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array}\right] R_3 \to R_3 - R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 2 & 0 & -1 & 1 \end{array}\right] \begin{array}{c} R_1 \to R_1 - R_2 \\ R_3 \to \frac{R_3}{2} \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 2 & -1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} \end{array}\right] R_1 \to R_1 - R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} \end{array}\right]$$

Example

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ by using Gauss-Jordan method.

Solution

$$[A|I_n] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array}\right] R_3 \to R_3 - R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 2 & 0 & -1 & 1 \end{array}\right] \begin{array}{c} R_1 \to R_1 - R_2 \\ R_3 \to \frac{R_3}{2} \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 2 & -1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} \end{array}\right] R_1 \to R_1 - R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} \end{array}\right] = [I_n|A^{-1}]$$

Example

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ by using Gauss-Jordan method.

Solution

$$[A|I_n] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \begin{array}{c} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array}\right] R_3 \to R_3 - R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 2 & 0 & -1 & 1 \end{array}\right] \begin{array}{c} R_1 \to R_1 - R_2 \\ R_3 \to \frac{R_3}{2} \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 2 & -1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} \end{array}\right] R_1 \to R_1 - R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} \end{array}\right] = [I_n|A^{-1}] \Longrightarrow \mathbf{A}^{-1} = \begin{bmatrix} 2 & -\frac{1}{2} & -\frac{1}{2} \\ -1 & 1 & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Exercise 12:

Using Gauss-Jordan elimination method find the inverse of

1. $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ Ans: $A^{-1} = \dfrac{1}{5}\begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$
2. $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 5 & -8 \\ -3 & -5 & 8 \end{bmatrix}$ Ans: $A^{-1} = \begin{bmatrix} 0 & -1 & -1 \\ 8 & -1 & 2 \\ 5 & -1 & -1 \end{bmatrix}$
3. $A = \begin{bmatrix} 3 & 2 & 3 & 1 \\ -2 & -1 & -1 & 0 \\ 3 & 2 & 4 & 2 \\ 3 & 2 & 3 & 2 \end{bmatrix}$ Ans: $A^{-1} = \begin{bmatrix} -2 & -2 & 1 & 0 \\ 4 & 3 & -3 & 1 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}$
4. $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 4 & 2 \end{bmatrix}$ Ans: $A^{-1} = \begin{bmatrix} 2 & 0 & -1 \\ 3 & -1 & -1 \\ -7 & 2 & 3 \end{bmatrix}$
5. $A = \begin{bmatrix} 2 & -3 & 5 \\ 0 & 3 & -2 \\ 1 & -2 & -2 \end{bmatrix}$ Ans: $A^{-1} = \begin{bmatrix} \frac{10}{29} & \frac{16}{29} & \frac{9}{29} \\ \frac{2}{29} & \frac{9}{29} & -\frac{4}{29} \\ \frac{3}{29} & -\frac{1}{29} & -\frac{6}{29} \end{bmatrix}$

6. $A = \begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix}$ Ans: $A^{-1} = \frac{1}{4}\begin{bmatrix} 27 & -11 & 3 \\ -11 & 5 & -1 \\ 3 & -1 & 1 \end{bmatrix}$

7. $A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ Ans: $A^{-1} = \begin{bmatrix} -1 & 2 & -4 \\ 1 & -1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$

Solution of (1)

$$[A|I_3] = \left[\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 2R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array}\right] R_3 \to R$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & -3 & -2 & -2 & 1 & 0 \\ 0 & -2 & -3 & -2 & 0 & 1 \end{array}\right] R_2 \to \frac{R_2}{-3} \left[\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & 1 & \frac{2}{3} & \frac{2}{3} & \frac{-1}{3} & 0 \\ 0 & -2 & -3 & -2 & 0 & 1 \end{array}\right] \begin{array}{l} R_1 \to R_1 - \\ R_3 \to R_3 + \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 1 & \frac{2}{3} & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & -\frac{5}{3} & -\frac{2}{3} & -\frac{2}{3} & 1 \end{array}\right] R_3 \to -\frac{3}{5}R_3 \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 1 & \frac{2}{3} & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & \frac{2}{5} & \frac{2}{5} & -\frac{3}{5} \end{array}\right] \begin{array}{l} R_1 \to R_1 - \frac{2}{3}R \\ R_2 \to R_2 - \frac{2}{3}R \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{3}{5} & \frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 0 & \frac{2}{5} & -\frac{3}{5} & \frac{2}{5} \\ 0 & 0 & 1 & \frac{2}{5} & \frac{2}{5} & -\frac{3}{5} \end{array}\right] = [I_3|A^{-1}] \Longrightarrow \mathbf{A^{-1}} = \begin{bmatrix} -\frac{3}{5} & \frac{2}{5} & \frac{2}{5} \\ \frac{2}{5} & -\frac{3}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{2}{5} & -\frac{3}{5} \end{bmatrix}$$

Definition(Elementary matrices)

A matrix obtained from identity matrix by applying elementary row operations is called an elementary matrix.

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Example

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_1 \longleftrightarrow R_3 \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = E_{1 \leftrightarrow 3}$$

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$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_2 \to \alpha R_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_{\alpha R_2}$$

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A matrix obtained from identity matrix by applying elementary row operations is called an elementary matrix.

Example

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$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_2 \rightarrow \alpha R_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_{\alpha R_2}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_2 \rightarrow R_2 + \alpha R_1 \begin{bmatrix} 1 & 0 & 0 \\ \alpha & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_{R_2+\alpha R_1}$$

Theorem

Applying an elementary row operation on any matrix is equivalent to pre-multiply that matrix by corresponding elementary matrix.

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Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} R_1 \longleftrightarrow R_2 \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix} = B$$

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Now let us compute corresponding elementary matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_1 \longleftrightarrow R_2 \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_{1\leftrightarrow 2}$$

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Applying an elementary row operation on any matrix is equivalent to pre-multiply that matrix by corresponding elementary matrix.

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} R_1 \longleftrightarrow R_2 \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix} = B$$

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$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_1 \longleftrightarrow R_2 \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_{1\leftrightarrow 2}$$

Then

$$E_{1\leftrightarrow 2}A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix} = B$$

Theorem

Inverse of an elementary matrix is again an elementary matrix.

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Example

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_1 \longleftrightarrow R_2 \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & \\ 0 & 0 & 1 \end{bmatrix} = E_{1\leftrightarrow 2} \implies (E_{1\leftrightarrow 2})^{-1} = E_{1\leftrightarrow 2}$$

Theorem

Inverse of an elementary matrix is again an elementary matrix.

Example

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_1 \longleftrightarrow R_2 \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & \\ 0 & 0 & 1 \end{bmatrix} = E_{1\leftrightarrow 2} \implies (E_{1\leftrightarrow 2})^{-1} = E_{1\leftrightarrow 2}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_2 \rightarrow \alpha R_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_{\alpha R_2} \implies (E_{\alpha R_2})^{-1} = E_{\frac{R_2}{\alpha}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\alpha} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Theorem

Inverse of an elementary matrix is again an elementary matrix.

Example

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_1 \longleftrightarrow R_2 \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & \\ 0 & 0 & 1 \end{bmatrix} = E_{1\leftrightarrow 2} \implies (E_{1\leftrightarrow 2})^{-1} = E_{1\leftrightarrow 2}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_2 \to \alpha R_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_{\alpha R_2} \implies (E_{\alpha R_2})^{-1} = E_{\frac{R_2}{\alpha}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\alpha} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_2 \to R_2 + \alpha R_1 \begin{bmatrix} 1 & 0 & 0 \\ \alpha & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_{R_2+\alpha R_1} \implies (E_{R_2+\alpha R_1})^{-1} = E_{R_2-\alpha R_1}$$

LDU factorization

Definition(LDU factorization)

A matrix A is said to have LDU factorization if

$$A = LDU$$

where

1. L is a lower triangular matrix with all main diagonal entries 1,
2. U is upper triangular matrix with all diagonal entries 1, and
3. D is a diagonal matrix with all main diagonal entries non-zero.

Steps to find *LDU* factorization of a matrix

Let A be square matrix. The following steps are required to find LDU factorization of a matrix.

Step-1 Get an echelon form of the matrix A by applying type-III elementary row operations on A.

Step-2 Factor out the main diagonal of matrix obtained in step-1 to get (**D**) and (**U**).

Step-3 Keep track of the elementary row operations (to get L through identity matrix).

Remark

If application of m elementary row operations of type-III converts A into Echelon form E and let $E_1, E_2, \dots E_m$ be the corresponding elementary matrices, i.e.,

$$E_m \cdots E_2 E_1 A = E = \begin{bmatrix} e_{11} & e_{12} & \cdots & e_{1n} \\ 0 & e_{22} & \cdots & e_{2n} \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \cdots & e_{nn} \end{bmatrix}$$

Then $D = \begin{bmatrix} e_{11} & 0 & \cdots & 0 \\ 0 & e_{22} & \cdots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \cdots & e_{nn} \end{bmatrix}$, $U = \begin{bmatrix} 1 & \frac{e_{12}}{e_{11}} & \cdots & \frac{e_{1n}}{e_{11}} \\ 0 & 1 & \cdots & \frac{e_{2n}}{e_{22}} \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$ and $L = E_1{}^{-1} \cdots E_M{}^{-1}$.

Example

Find the LDU factorization of the matrix $A = \begin{bmatrix} 3 & 9 \\ 15 & 49 \end{bmatrix}$.

Example

Find the LDU factorization of the matrix $A = \begin{bmatrix} 3 & 9 \\ 15 & 49 \end{bmatrix}$.

Solution

$$A = \begin{bmatrix} 3 & 9 \\ 15 & 49 \end{bmatrix} R_2 \to R_2 - 5R_1 \begin{bmatrix} 3 & 9 \\ 0 & 4 \end{bmatrix}$$

Thus, $D = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$ *and* $U = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. *Again,*

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} R_2 \to R_2 + 5R_1 \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix} = L$$

Then $A = LDU = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$ *(Verify!).*

Example

Find the LDU factorization of the matrix $A = \begin{bmatrix} 4 & -20 & -12 \\ -8 & 45 & 44 \\ 20 & -105 & -79 \end{bmatrix}$

Example

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Solution

$$A = \begin{bmatrix} 4 & -20 & -12 \\ -8 & 45 & 44 \\ 20 & -105 & -79 \end{bmatrix} \xrightarrow[R_3 \to R_3 - 5R_1]{R_2 \to R_2 + 2R_1} \begin{bmatrix} 4 & -20 & -12 \\ 0 & 5 & 20 \\ 0 & -5 & -19 \end{bmatrix} \xrightarrow{R_3 \to R_3 + R_2} \begin{bmatrix} 4 & -20 & -12 \\ 0 & 5 & 20 \\ 0 & 0 & 1 \end{bmatrix}$$

Thus, $D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ *and* $U = \begin{bmatrix} 1 & -5 & -3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$. *Also,*

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow[R_3 \to R_3 + 5R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & -1 & 1 \end{bmatrix} = L$$

Therefore, $A = LDU$ *(Verify!).*

Example

Transform the given matrix into LDU factorization form $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$.

Example

Transform the given matrix into LDU factorization form $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$.

Solution

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} R_2 \to R_2 + \frac{1}{2} R_1 \begin{bmatrix} 2 & -1 & 0 \\ 0 & \frac{3}{2} & -1 \\ 0 & -1 & 2 \end{bmatrix} R_3 \to R_3 + \frac{2}{3} R_2 \begin{bmatrix} 2 & -1 & 0 \\ 0 & \frac{3}{2} & -1 \\ 0 & 0 & \frac{4}{3} \end{bmatrix}$$

Thus, $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & \frac{4}{3} \end{bmatrix}$ and $U = \begin{bmatrix} 1 & \frac{-1}{2} & 0 \\ 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 1 \end{bmatrix}$. Also,

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_3 \to R_3 - \frac{2}{3} R_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix} R_2 \to R_2 - \frac{1}{2} R_1 \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix} = L$$

Therefore $A = LDU$.

Exercise 13:

Find the LDU factorization of the following matrices.

1. $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ **Ans:** $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $U = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$
2. $A = \begin{bmatrix} 2 & 4 & -16 \\ -2 & -8 & -10 \\ -6 & -32 & -66 \end{bmatrix}$

   Ans: $L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -3 & 5 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 16 \end{bmatrix}$, $U = \begin{bmatrix} 1 & 2 & -8 \\ 0 & 1 & \frac{13}{2} \\ 0 & 0 & 1 \end{bmatrix}$
3. $A = \begin{bmatrix} 2 & 1 \\ -4 & -6 \end{bmatrix}$
4. $A = \begin{bmatrix} 2 & 1 & -4 \\ 2 & 2 & -2 \\ 6 & 3 & -11 \end{bmatrix}$

Exercise 14:

Find the LDU factorization of the matrix $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 8 & 5 \\ 1 & 11 & 4 \end{bmatrix}$.

Solution

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 8 & 5 \\ 1 & 11 & 4 \end{bmatrix} \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - R_1 \end{matrix} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 2 & 1 \\ 0 & 8 & 2 \end{bmatrix} R_3 \to R_3 - 4R_2 \begin{bmatrix} 1 & 3 & 2 \\ 0 & 2 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

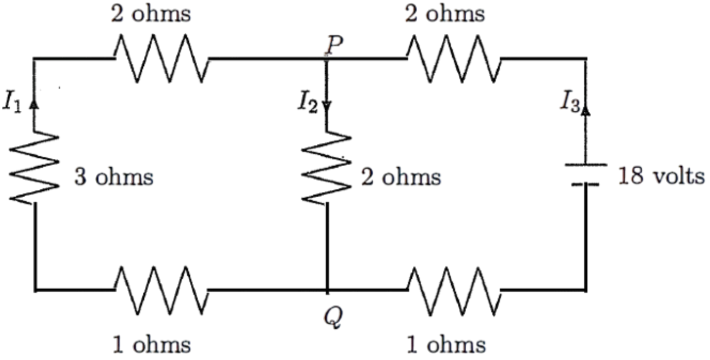
Thus, $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ and $U = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$. Also,

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_3 \to R_3 + 4R_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix} \begin{matrix} R_3 \to R_3 + R_1 \\ R_2 \to R_2 + 2R_1 \end{matrix} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 4 & 1 \end{bmatrix} = L$$

Therefore, $A = LDU$

Application of linear equations in electrical network

In an electrical network, a simple current flow may be illustrated by a diagram like the one below. Such a network involves only voltage sources, like batteries, and resistors, like bulbs, motors, or refrigerators. The voltage is measured in volts, the resistance in ohms, and the current flow in amperes (amps, in short).



For such an electrical network, current flow is governed by the following three laws:

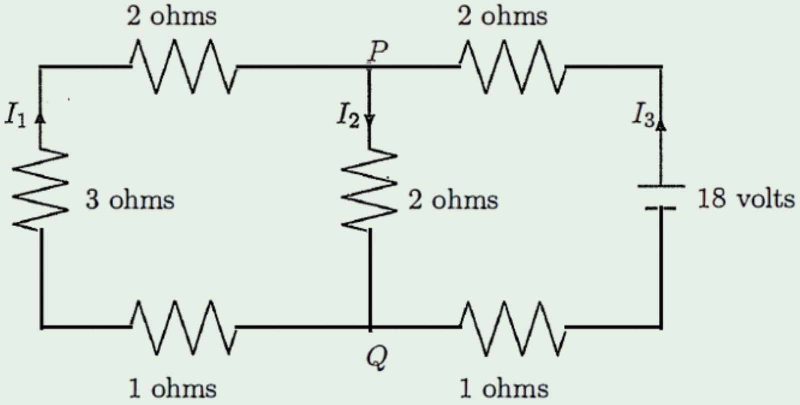
1. **Ohm's Law:** The voltage drop V across a resistor is the product of the current I and the resistance R:
$$V = IR.$$

2. **Kirchhoff's Current Law (KCL):** The current flow into a node equals the current flow out of the node.

3. **Kirchhoff's Voltage Law (KVL):** The algebraic sum of the voltage drops around a closed loop equals the total voltage sources in the loop.

Example

Determine the currents in the network given in the figure below.



Solution

By applying KCL to nodes P and Q, we get equations

$$I_1 + I_3 = I_2 \textit{ at } P,$$
$$I_2 = I_1 + I3 \textit{ at } Q.$$

Observe that both equations are the same. By applying KVL to each of the loops in the network clockwise direction, we get

$$6I_1 + 2I_2 = 0 \textit{ from the left loop}$$
$$2I_2 + 3I_3 = 18 \textit{ from the right loop}$$

Collecting all the equations, we get the following system of linear equations

$$I_1 - I_2 + I_3 = 0$$
$$6I_1 + 2I_2 = 0$$
$$2I_2 + 3I_3 = 18$$

Solution cont...

The augmented matrix of above systme of linear equation is

$$A^* = \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 6 & 2 & 0 & 0 \\ 0 & 2 & 3 & 18 \end{array}\right]$$

The RREF of above matrix is

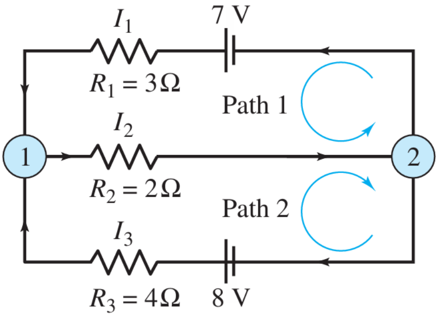
$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

From here, $\text{rank}(A) = \text{rank}(A^*) = \#$ of variables . Thus, system is consistent and has a unique solution. The above RREF gives the following solution

$$I_1 = -1,\ I_2 = 3,\ I_3 = 4$$

Exercise 15:

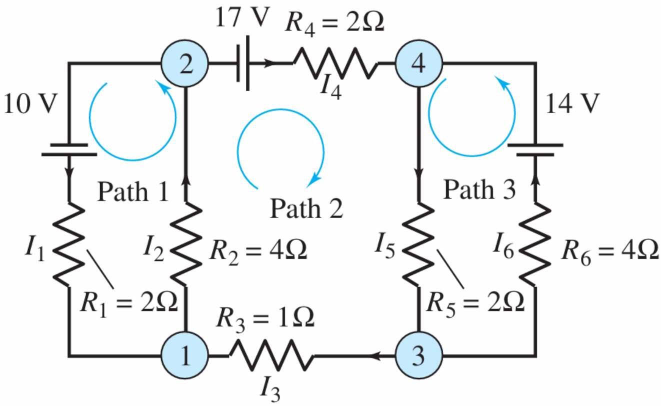
Determine the currents I_1, I_2 and I_3 for the electrical network shown in Figure below



Ans: $I_1 = 1$, $I_2 = 2$ and $I_3 = 1$.

Exercise 16:

Determine the currents I_1, I_2, I_3, I_4, I_5 and I_6 for the electrical network shown in Figure below.

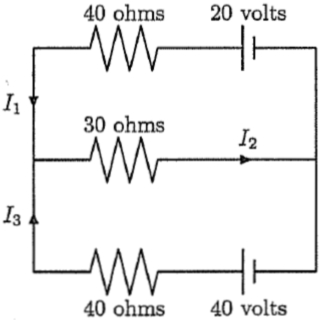


Ans: $I_1 = 1$, $I_2 = 2$, $I_3 = 1$, $I_4 = 1$, $I_5 = 3$ and $I_6 = 2$.

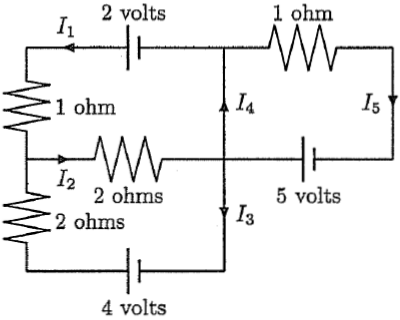
Exercise 17:

Determine the currents in the following networks.

(1)

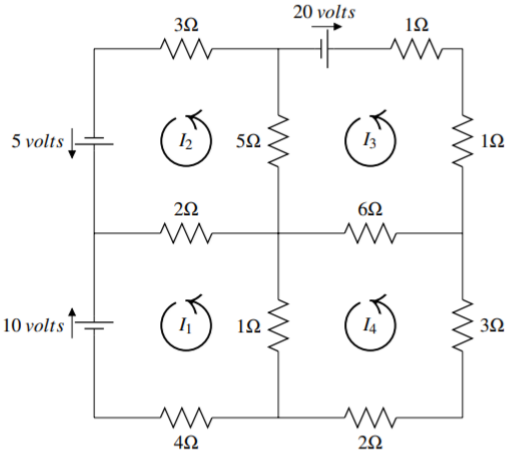


(2)



Exercise 18:

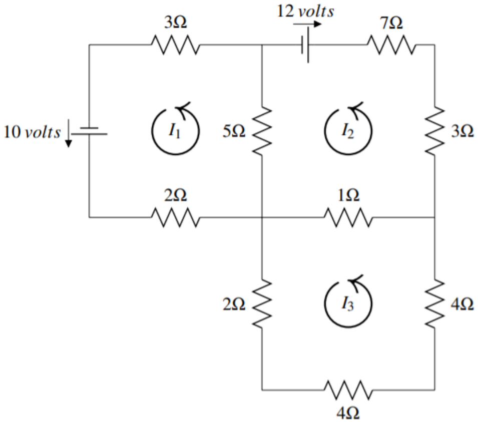
Consider the following diagram of four circuits.



Write equations for each circuits and then give a solution to the resulting system of linear equations. **Ans:** $I_1 = -\frac{750}{373}$, $I_2 = -\frac{1421}{1119}$, $I_3 = -\frac{3061}{1119}$, $I_4 = -\frac{1718}{1119}$

Exercise 19:

Consider the following diagram of three circuits.

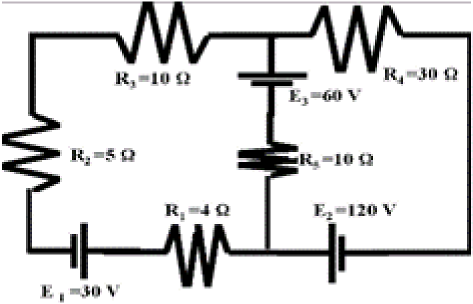


The current in amps in the four circuits is denoted by I_1, I_2, I_3 and it is understood that the motion is in the counter clockwise direction. If I_k ends up being negative, then it just means the current flows in the clockwise direction. Find I_1, I_2, I_3.

Ans: $I_1 = \frac{218}{295}, I_2 = -\frac{154}{295}, I_3 = -\frac{14}{295}$

Exercise 20:

Find the current in circuit for the following network



Application of linear equations in balancing chemical reactions

- The tools of linear algebra can also be used in the subject area of Chemistry, specifically for balancing chemical reactions.
  Consider the chemical reaction

  $$SnO_2 + H_2 \to Sn + H_2O$$

  Here the elements involved are tin (Sn), oxygen (O), and hydrogen (H). A chemical reaction occurs and the result is a combination of tin (Sn) and water (H_2O). When considering chemical reactions, we want to investigate how much of each element we began with and how much of each element is involved in the result.

- An important theory we will use here is the mass balance theory. It tells us that we cannot create or delete elements within a chemical reaction.

- An important theory we will use here is the mass balance theory. It tells us that we cannot create or delete elements within a chemical reaction.

- For example, in the above expression, we must have the same number of oxygen, tin, and hydrogen on both sides of the reaction. Notice that this is not currently the case. For example, there are two oxygen atoms on the left and only one on the right. In order to fix this, we want to find numbers x, y, z, w such that

$$xSnO_2 + yH_2 \to zSn + wH_2O$$

where both sides of the reaction have the same number of atoms of the various elements.

❋ This is a familiar problem. We can solve it by setting up a system of equations in the variables x, y, z, w. Thus you need

$$\begin{aligned} Sn: &\quad x = z \\ O: &\quad 2x = w \\ H: &\quad 2y = 2w \end{aligned}$$

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$$\begin{aligned} Sn: &\quad x - z = 0 \\ O: &\quad 2x - w = 0 \\ H: &\quad 2y - 2w = 0 \end{aligned}$$

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- The augmented matrix for this system of equations is given by

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 0 \\ 2 & 0 & 0 & -1 & 0 \\ 0 & 2 & 0 & -2 & 0 \end{array}\right]$$

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$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 0 \\ 2 & 0 & 0 & -1 & 0 \\ 0 & 2 & 0 & -2 & 0 \end{array}\right]$$

- The reduced row-echelon form of this matrix is

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 \end{array}\right]$$

❋ The solution is given by

$$\begin{aligned} x - \tfrac{1}{2}w &= 0 \\ y - w &= 0 \\ z - \tfrac{1}{2}w &= 0 \end{aligned}$$

which we can write as

$$\begin{aligned} x &= \tfrac{1}{2}t \\ y &= t \\ z &= \tfrac{1}{2}t \\ w &= t \end{aligned}$$

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❃ For example, let $w = 2$ and this would yield $x = 1, y = 2$, and $z = 1$. We can put these values back into the expression for the reaction which yields

$$SnO_2 + 2H_2 \rightarrow Sn + 2H_2O$$

Observe that each side of the expression contains the same number of atoms of each element. This means that it preserves the total number of atoms, as required, and so the chemical reaction is balanced.

Example

Potassium is denoted by K, oxygen by O, phosphorus by P and hydrogen by H. Consider the reaction given by

$$KOH + H_3PO_4 \rightarrow K_3PO_4 + H_2$$

Balance this chemical reaction.

Solution

We need to find values for x,y,z,w such that

$$xKOH + yH_3PO_4 \rightarrow zK_3PO_4 + wH_2O$$

preserves the total number of atoms of each element.
Finding these values can be done by finding the solution to the following system of equations.

$$\begin{array}{ll} K: & x = 3z \\ O: & x + 4y = 4z + w \\ H: & x + 3y = 2w \\ P: & y = z \end{array}$$

The augmented matrix for this system is

$$\left[\begin{array}{rrrr|r} 1 & 0 & -3 & 0 & 0 \\ 1 & 4 & -4 & -1 & 0 \\ 1 & 3 & 0 & -2 & 0 \\ 0 & 1 & -1 & 0 & 0 \end{array}\right]$$

and the reduced row-echelon form is

$$\left[\begin{array}{cccr|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & -\frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The solution is given by

$$\begin{gathered} x - w = 0 \\ y - \frac{1}{3}w = 0 \\ z - \frac{1}{3}w = 0 \end{gathered} \qquad \text{which can be written as} \qquad \begin{gathered} x = t \\ y = \frac{1}{3}t \\ z = \frac{1}{3}t \\ w = t \end{gathered}$$

Choose a value for t, say 3. Then $w = 3$ and this yields $x = 3, y = 1, z = 1$. It follows that the balanced reaction is given by

$$3KOH + 1H_3PO_4 \rightarrow 1K_3PO_4 + 3H_2O$$

Note that this results in the same number of atoms on both sides.

and the reduced row-echelon form is

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & -\frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The solution is given by

$$\begin{aligned} x - w &= 0 \\ y - \tfrac{1}{3}w &= 0 \\ z - \tfrac{1}{3}w &= 0 \end{aligned} \qquad \text{which can be written as} \qquad \begin{aligned} x &= t \\ y &= \tfrac{1}{3}t \\ z &= \tfrac{1}{3}t \\ w &= t \end{aligned}$$

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$$3KOH + 1H_3PO_4 \rightarrow 1K_3PO_4 + 3H_2O$$

Note that this results in the same number of atoms on both sides.

Remark

Of course these numbers you are finding would typically be the number of moles of the molecules on each side. Thus three moles of KOH added to one mole of H_3PO_4 yields one mole of K_3PO_4 and three moles of H_2O.

Exercise 21:

Consider the following reaction of sugar fermentation.

$$C_6H_{12}O_6 \rightarrow CO_2 + C_2H_5OH$$

Balance this chemical reaction. **Ans:** $C_6H_{12}C_6 \rightarrow 2CO_2 + 2C_2H_5OH$.

Exercise 22:

Propane (C_3H_8) burns in oxygen to form carbon dioxide and steam (propane combustion reaction).

$$C_3H_8 + O_2 \rightarrow CO_2 + H_2O$$

Balance this chemical reaction. **Ans:** $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$.

Exercise 23:

Consider the following reaction of photosynthesis.

$$CH_3COF + H_2O \rightarrow CH_3COOH + HF$$

Balance this chemical reaction. **Ans:** $CH_3COF + H_2O \rightarrow CH_3COOH + HF$.

Exercise 24:

Balance the following chemical reactions.

1. $KNO_3 + H_2CO_3 \rightarrow K_2CO_3 + HNO_3$
2. $AgI + Na_2S \rightarrow Ag_2S + NaI$
3. $Ba_3N_2 + H_2O \rightarrow Ba(OH)_2 + NH_3$
4. $CaCl_2 + Na_3PO_4 \rightarrow Ca_3(PO_4)_2 + NaCl$

Application of invertible matrices in cryptography

- Cryptography is the study of sending messages in disguised form (secret codes) so that only the intended recipients can remove the disguise and read the message; modern cryptography uses advanced mathematics.
- As another application of invertible matrices, we introduce a simple coding. Suppose we associate a prescribed number with every letter in the alphabet; for example,

$$\begin{array}{ccccccccccccccc}
\mathrm{A} & \mathrm{B} & \mathrm{C} & \mathrm{D} & \mathrm{E} & \mathrm{F} & \mathrm{G} & \mathrm{H} & \mathrm{I} & \mathrm{J} & \mathrm{K} & \mathrm{L} & \mathrm{M} & \mathrm{N} & \mathrm{O} \\
\updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow \\
0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 \\
\hline
\mathrm{P} & \mathrm{Q} & \mathrm{R} & \mathrm{S} & \mathrm{T} & \mathrm{U} & \mathrm{V} & \mathrm{W} & \mathrm{X} & \mathrm{Y} & \mathrm{Z} & \textit{Blank} & ? & ! & \\
\updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \\
15 & 16 & 17 & 18 & 19 & 20 & 21 & 22 & 23 & 24 & 25 & 26 & 27 & 28 &
\end{array}$$

- Suppose that we want to send the message "GOOD LUCK" and we want to use a 3×3 invertible matrix to decode this message. We divide the letters of the message into groups of three.

GOO D-L UCK

❋ Now we assign the numbers their corresponding letters from the table, and convert each triplet of numbers into 3×1 matrices. We get

$$\begin{bmatrix} G \\ O \\ O \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \\ 14 \end{bmatrix}, \quad \begin{bmatrix} D \\ - \\ L \end{bmatrix} = \begin{bmatrix} 3 \\ 26 \\ 11 \end{bmatrix}, \begin{bmatrix} U \\ C \\ K \end{bmatrix} = \begin{bmatrix} 20 \\ 2 \\ 10 \end{bmatrix}$$

❋ Next, choose a invertible matrix 3×3 matrix A, say

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

which is supposed to be known to both sender and receiver. Then A translates our message into

$$A \begin{bmatrix} 6 \\ 14 \\ 14 \end{bmatrix} = \begin{bmatrix} 6 \\ 26 \\ 34 \end{bmatrix}, A \begin{bmatrix} 3 \\ 26 \\ 11 \end{bmatrix} = \begin{bmatrix} 3 \\ 32 \\ 40 \end{bmatrix}, A \begin{bmatrix} 20 \\ 2 \\ 10 \end{bmatrix} = \begin{bmatrix} 20 \\ 42 \\ 32 \end{bmatrix}$$

❋ By putting the components of the resulting vectors consecutively, we transmit

$$6, 26, 34, 3, 32, 40, 20, 42, 32. \qquad \text{(Encoded message)}$$

❋ To decode this message, the receiver may follow the following process.
To decode it, first break the message into three vectors in $\mathbb{R}^3$ as before:

$$\begin{bmatrix} 6 \\ 26 \\ 34 \end{bmatrix}, \begin{bmatrix} 3 \\ 32 \\ 40 \end{bmatrix}, \begin{bmatrix} 20 \\ 42 \\ 32 \end{bmatrix}$$

❋ We want to find three vectors x_1, x_2, x_3 in $\mathbb{R}^3$ such that

$$Ax_1 = \begin{bmatrix} 6 \\ 26 \\ 34 \end{bmatrix}, Ax_2 = \begin{bmatrix} 3 \\ 32 \\ 40 \end{bmatrix}, Ax_3 = \begin{bmatrix} 20 \\ 42 \\ 32 \end{bmatrix}$$

❋ Note that $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$. This implies

$$x_1 = A^{-1}\begin{bmatrix} 6 \\ 26 \\ 34 \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \\ 14 \end{bmatrix}, x_2 = A^{-1}\begin{bmatrix} 3 \\ 32 \\ 40 \end{bmatrix} = \begin{bmatrix} 3 \\ 26 \\ 11 \end{bmatrix}, x_3 = A^{-1}\begin{bmatrix} 20 \\ 42 \\ 32 \end{bmatrix} = \begin{bmatrix} 20 \\ 2 \\ 10 \end{bmatrix}$$

❋ The numbers one obtains are

6, 14, 14, 3, 26, 11, 20, 2, 10

Using our correspondence between letters and numbers, the message we have received is

GOOD LUCK (Decoded message)

Remark

We summarize:

TO ENCODE A MESSAGE

1. Divide the letters of the message into groups of two or three.
2. Convert each group into a string of numbers by assigning a number to each letter of the message. Remember to assign letters to blank spaces.
3. Convert each group of numbers into column matrices.
4. Convert these column matrices into a new set of column matrices by multiplying them with a compatible square matrix of your choice that has an inverse. This new set of numbers or matrices represents the coded message.

TO DECODE A MESSAGE

1. Take the string of coded numbers and multiply it by the inverse of the matrix that was used to encode the message.
2. Associate the numbers with their corresponding letters.

Example

Decode the following message that was encoded using the matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$.

$$\begin{bmatrix} 21 \\ 26 \end{bmatrix}, \begin{bmatrix} 37 \\ 53 \end{bmatrix}, \begin{bmatrix} 45 \\ 54 \end{bmatrix}, \begin{bmatrix} 74 \\ 101 \end{bmatrix}, \begin{bmatrix} 53 \\ 69 \end{bmatrix}$$

using the correspondence shown below

$$\begin{array}{ccccccccccccccc}
\mathrm{A} & \mathrm{B} & \mathrm{C} & \mathrm{D} & \mathrm{E} & \mathrm{F} & \mathrm{G} & \mathrm{H} & \mathrm{I} & \mathrm{J} & \mathrm{K} & \mathrm{L} & \mathrm{M} & \mathrm{N} & \mathrm{O} \\
\updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow \\
1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 \\
\hline
\mathrm{P} & \mathrm{Q} & \mathrm{R} & \mathrm{S} & \mathrm{T} & \mathrm{U} & \mathrm{V} & \mathrm{W} & \mathrm{X} & \mathrm{Y} & \mathrm{Z} & \textit{Blank} & ? & ! & \\
\updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \\
16 & 17 & 18 & 19 & 20 & 21 & 22 & 23 & 24 & 25 & 26 & 27 & 28 & 29 &
\end{array}$$

Solution

Note that $A^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$. Then

$$A^{-1}\begin{bmatrix} 21 \\ 26 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 21 \\ 26 \end{bmatrix} = \begin{bmatrix} 11 \\ 5 \end{bmatrix}$$

$$A^{-1}\begin{bmatrix} 37 \\ 53 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 37 \\ 53 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$$

$$A^{-1}\begin{bmatrix} 45 \\ 54 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 45 \\ 54 \end{bmatrix} = \begin{bmatrix} 27 \\ 9 \end{bmatrix}$$

$$A^{-1}\begin{bmatrix} 74 \\ 101 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 74 \\ 101 \end{bmatrix} = \begin{bmatrix} 20 \\ 27 \end{bmatrix}$$

$$A^{-1}\begin{bmatrix} 53 \\ 69 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 53 \\ 69 \end{bmatrix} = \begin{bmatrix} 21 \\ 16 \end{bmatrix}$$

The numbers one obtains are

$$11,\ 5,\ 5,\ 16,\ 27,\ 9,\ 20,\ 27,\ 21,\ 16$$

Using given correspondence between letters and numbers, the decoded message is **KEEP IT UP**.

Exercise 25:

Encode "TAKE UFO" using the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.

Exercise 26:

Use matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ to encode the message ATTACK NOW!

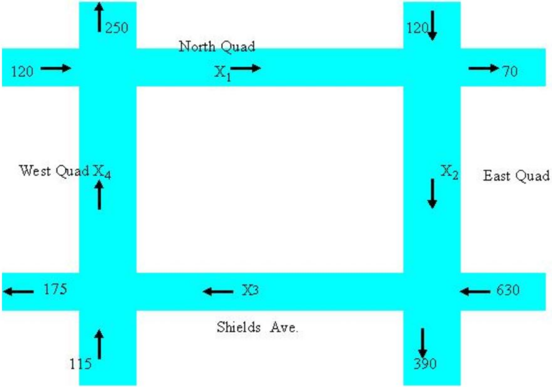
Exercise 27:

Using the matrix $B = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$, encode the message: ATTACK NOW!

Application of system of linear equations in traffic flow

Example

The following diagram shows part of the central section of UC Davis campus. Assume that the streets are one way, and that the average number of bikes entering and leaving this section during the 10 minutes breaks between classes is given in the chart. Find the amount of the traffic between each of four intersection.



Solution

Assume that the number of bikes entering each intersection is equal to the number of the bikes leaving the intersection. For each intersection, this fact can be shown by an equation.

$$x_4 + 120 = x_1 + 250$$
$$x_3 + 115 = x_4 + 175$$
$$x_2 + 630 = x_3 + 390$$
$$x_1 + 120 = x_2 + 70$$

Eigenvalues and Eigenvectors

Definition (Eigenvalue and Eigenvector)

- Let A be a square matrix of size $n \times n$. A number λ is called an **eigenvalue** of A if there exists a non-zero vector $\vec{v}$ such that

$$A\vec{v} = \lambda\vec{v}.$$

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- The set of all eigenvectors associated with the eigenvalue λ forms a subspace, and is called the **eigenspace** associated with λ and it is denoted by E_λ.

- The pair $(\lambda, \vec{v})$ is called an eigen pair of A.

Example

Let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$. Then

$$A \begin{bmatrix} 6 \\ -5 \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4 \begin{bmatrix} 6 \\ -5 \end{bmatrix}$$

Thus, -4 is an eigenvalue of A and $\begin{bmatrix} 6 \\ -5 \end{bmatrix}$ is corresponding eigenvector.

Computation of eigenvalues

* Let A be a square matrix of size $n \times n$ and $(\lambda, \vec{v})$ be an eigen pair of A. Then

$$A\vec{v} = \lambda\vec{v} \implies (A - \lambda I_n)\vec{v} = 0 \tag{1}$$

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- Note that $\det(A - \lambda I_n)$ is a polynomial of degree n and it is called Characteristic Polynomial and is denoted by $p_A(\lambda)$. Thus the eigenvalues of A are the roots of the characteristic polynomial.

Example

Let $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$. Then the characteristic polynomial of A is

$$p_A(\lambda) = \det(A - \lambda I_2) = \det \begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix}$$
$$= \lambda^2 + 4\lambda - 21 = (\lambda + 7)(\lambda - 3),$$

whose roots are -7 and 3, and these are the eigenvalues of A.

Computation of eigenvectors

From (1), we see that eigenvectors corresponding to the eigenvalue λ are the solutions of the homogeneous system of linear equations $(A - \lambda I_n)X = 0$.

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Procedure

- Find the eigenvalues of A.
- For each eigenvalue λ, solve the linear system $(A - \lambda I_n)X = 0$.
- The set of all solutions of this linear system is the eigenspace E_λ of all eigenvectors of A with eigenvalue λ.

Example

Let $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$. Then, we know that eigenvalues of A are $\lambda = -7$ and $\lambda = 3$. Now for $\lambda = -7$, we have

$$\begin{aligned} (A - \lambda I_2)X = 0 &\implies (A + 7I_2)X = 0 \\ &\implies \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ &\implies 3x_1 + x_2 = 0 \end{aligned}$$

Thus,

$$\begin{aligned} E_{-7} &= \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \middle| \; 3x_1 + x_2 = 0 \right\} \\ &= \left\{ \begin{bmatrix} x_1 \\ -3x_1 \end{bmatrix} \middle| \; x_1 \in \mathbb{R} \right\} \\ &= \left\{ x_1 \begin{bmatrix} 1 \\ -3 \end{bmatrix} \middle| \; x_1 \in \mathbb{R} \right\} \end{aligned}$$

So, an eigenvector corresponding to $\lambda = -7$ is $\begin{bmatrix} 1 \\ -3 \end{bmatrix}$. For $\lambda = 3$, we get

$$\begin{aligned}
(A - \lambda I_2)X = 0 &\implies (A - 3I_2)X = 0 \\
&\implies \begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\
&\implies -x_1 + 3x_2 = 0
\end{aligned}$$

Thus,

$$\begin{aligned}
E_3 &= \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \middle| -x_1 + 3x_2 = 0 \right\} \\
&= \left\{ \begin{bmatrix} 3x_2 \\ x_2 \end{bmatrix} \middle| x_2 \in \mathbb{R} \right\} \\
&= \left\{ x_2 \begin{bmatrix} 3 \\ 1 \end{bmatrix} \middle| x_2 \in \mathbb{R} \right\}
\end{aligned}$$

So, an eigenvector corresponding to $\lambda = 3$ is $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$.

Example

For the following matrix, compute the characteristic polynomial, eigenvalue, and eigenvectors corresponding to each eigenvalue.

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

Solution

Characteristic polynomial

$$p_A(\lambda) = \det(A - \lambda I_3) \implies \det \begin{bmatrix} 1-\lambda & 1 & -1 \\ 0 & 2-\lambda & 0 \\ -1 & 1 & 1-\lambda \end{bmatrix} = \lambda(2-\lambda)^2$$

Solution

Characteristic polynomial

$$p_A(\lambda) = \det(A - \lambda I_3) \implies \det \begin{bmatrix} 1-\lambda & 1 & -1 \\ 0 & 2-\lambda & 0 \\ -1 & 1 & 1-\lambda \end{bmatrix} = \lambda(2-\lambda)^2$$

Hence, the eigenvalues are $\lambda_1 = \lambda_2 = 2$ and $\lambda_3 = 0$. For eigenvectors corresponding to $\lambda_1 = \lambda_2 = 2$, we compute

$$(A - 2I_3)X = 0 \implies \begin{bmatrix} -1 & 1 & -1 \\ 0 & 0 & 0 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \implies -x_1 + x_2 - x_3 = 0$$

Solution

Characteristic polynomial

$$p_A(\lambda) = \det(A - \lambda I_3) \implies \det \begin{bmatrix} 1-\lambda & 1 & -1 \\ 0 & 2-\lambda & 0 \\ -1 & 1 & 1-\lambda \end{bmatrix} = \lambda(2-\lambda)^2$$

Hence, the eigenvalues are $\lambda_1 = \lambda_2 = 2$ and $\lambda_3 = 0$. For eigenvectors corresponding to $\lambda_1 = \lambda_2 = 2$, we compute

$$(A - 2I_3)X = 0 \implies \begin{bmatrix} -1 & 1 & -1 \\ 0 & 0 & 0 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \implies -x_1 + x_2 - x_3 = 0$$

from which it follows that

$$E_2 = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : -x_1 + x_2 - x_3 = 0 \right\} = \left\{ \begin{bmatrix} x_2 - x_3 \\ x_2 \\ x_3 \end{bmatrix} : x_2, x_3 \in \mathbb{R} \right\}$$
$$= \left\{ x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} : x_2, x_3 \in \mathbb{R} \right\}$$

Thus, there are two distinct eigenvectors $\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}-1\\0\\1\end{bmatrix}$ corresponding to $\lambda = 2$.

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For eigenvector corresponding to $\lambda_3 = 0$, we compute

$$(A - 0I_3)X = 0 \implies AX = 0 \implies \begin{bmatrix}1 & 1 & -1\\0 & 2 & 0\\-1 & 1 & 1\end{bmatrix} X = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

Thus, there are two distinct eigenvectors $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ corresponding to $\lambda = 2$.

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The augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 2 & 0 & 0 \\ -1 & 1 & 1 & 0 \end{array}\right] \begin{matrix} R_3 \to R_3 + R_1 \\ R_2 \to \frac{R_2}{2} \end{matrix} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right] R_3 \to R_3 - R_2 \left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

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from which it follows that $x_1 + x_2 - x_3 = 0$ and $x_2 = 0$, which gives $x_1 = x_3$. Thus,

$$E_0 = \left\{\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} \;:\; x_1 = x_3 \;\&\; x_2 = 0\right\} = \left\{\begin{bmatrix}x_3\\0\\x_3\end{bmatrix} \;:\; x_3 \in \mathbb{R}\right\} = \left\{x_3\begin{bmatrix}1\\0\\1\end{bmatrix} \;:\; x_3 \in \mathbb{R}\right\}.$$

Thus, there are two distinct eigenvectors $\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}-1\\0\\1\end{bmatrix}$ corresponding to $\lambda = 2$.

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$$E_0 = \left\{ \begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} : x_1 = x_3 \ \&\ x_2 = 0 \right\} = \left\{ \begin{bmatrix}x_3\\0\\x_3\end{bmatrix} : x_3 \in \mathbb{R} \right\} = \left\{ x_3 \begin{bmatrix}1\\0\\1\end{bmatrix} : x_3 \in \mathbb{R} \right\}.$$

Hence, $\begin{bmatrix}1\\0\\1\end{bmatrix}$ is the eigenvector corresponding to eigenvalue $\lambda = 0$.

Example

Find the eigenvalue and eigenvector of the following matrices

(a) $A = \begin{bmatrix} 3 & 0 & 0 \\ -3 & 4 & 9 \\ 0 & 0 & 3 \end{bmatrix}$

(b) $B = \begin{bmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{bmatrix}$

Solution

(a) The characteristics polynomial

$$p_A(\lambda) = \det(A - \lambda I_3) = \det \begin{bmatrix} 3-\lambda & 0 & 0 \\ -3 & 4-\lambda & 9 \\ 0 & 0 & 3-\lambda \end{bmatrix} = (3-\lambda)^2(4-\lambda)$$

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So, the eigenvalues of A are $\lambda = 3$ and $\lambda = 4$.

Eigenvector corresponding to eigenvalue $\lambda = 3$:

$$(A - 3I_3)X = 0$$

Consider Augmented matrix $\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ -3 & 1 & 9 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xRightarrow{\text{rref}} \left[\begin{array}{ccc|c} 1 & -\frac{1}{3} & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.

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(a) The characteristics polynomial

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So, from here, we get $x_1 - \frac{1}{3}x_2 - 3x_3 = 0$, i.e.,

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{3}x_2 + 3x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} \frac{1}{3} \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

Eigenvectors corresponding to $\lambda = 3$ are $v_1 = \begin{bmatrix} \frac{1}{3} \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$.

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$$(A - 4I_3)X = 0$$

Consider the Augmented matrix $\left[\begin{array}{ccc|c} -1 & 0 & 0 & 0 \\ -3 & 0 & 9 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.

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So, from here, we get $x_1 = 0$ and $x_3 = 0$, i.e.,

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} x_2.$$

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An eigenvector corresponding to $\lambda = 4$ is $v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

(b) Characteristics polynomial

$$p_B(\lambda) = \det(B - \lambda I) = \det \begin{bmatrix} 5-\lambda & 0 & 1 \\ 0 & -2-\lambda & 0 \\ 1 & 0 & 5-\lambda \end{bmatrix} = (-2-\lambda)(4-\lambda)(6-\lambda).$$

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So, the eigenvalues of B are $\lambda = -2$, $\lambda = 4$ and $\lambda = 6$.

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So, the eigenvalues of B are $\lambda = -2$, $\lambda = 4$ and $\lambda = 6$.

Eigenvector corresponding to eigenvalue $\lambda = -2$:

$$(A - 3I)X = 0$$

Consider the Augmented matrix $\left[\begin{array}{ccc|c} 7 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 7 & 0 \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.

(b) Characteristics polynomial

$$p_B(\lambda) = \det(B - \lambda I) = \det \begin{bmatrix} 5-\lambda & 0 & 1 \\ 0 & -2-\lambda & 0 \\ 1 & 0 & 5-\lambda \end{bmatrix} = (-2-\lambda)(4-\lambda)(6-\lambda).$$

So, the eigenvalues of B are $\lambda = -2$, $\lambda = 4$ and $\lambda = 6$.

Eigenvector corresponding to eigenvalue $\lambda = -2$:

$$(A - 3I)X = 0$$

Consider the Augmented matrix $\begin{bmatrix} 7 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 1 & 0 & 7 & | & 0 \end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$.

So, from here, we get $x_1 = 0$ and $x_3 = 0$. i.e.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

An eigenvector corresponding to $\lambda = -2$ is $v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

Eigenvector corresponding to eigenvalue $\lambda = 4$:

$$(A - 4I)X = 0$$

Consider the Augmented matrix $\begin{bmatrix} 1 & 0 & 1 & | & 0 \\ 0 & -6 & 0 & | & 0 \\ 1 & 0 & 1 & | & 0 \end{bmatrix} \xRightarrow{\text{rref}} \begin{bmatrix} 1 & 0 & 1 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$.

Eigenvector corresponding to eigenvalue $\lambda = 4$:

$$(A - 4I)X = 0$$

Consider the Augmented matrix $\begin{bmatrix} 1 & 0 & 1 & | & 0 \\ 0 & -6 & 0 & | & 0 \\ 1 & 0 & 1 & | & 0 \end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix} 1 & 0 & 1 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$.

So, from here, we get $x_1 + x_3 = 0$ and $x_2 = 0$. i.e.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ -x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Eigenvector corresponding to eigenvalue $\lambda = 4$:

$$(A - 4I)X = 0$$

Consider the Augmented matrix $\begin{bmatrix} 1 & 0 & 1 & | & 0 \\ 0 & -6 & 0 & | & 0 \\ 1 & 0 & 1 & | & 0 \end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix} 1 & 0 & 1 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$.

So, from here, we get $x_1 + x_3 = 0$ and $x_2 = 0$. i.e.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ -x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

An eigenvector corresponding to $\lambda = 4$ is $v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$.

Eigenvector corresponding to eigenvalue $\lambda = 6$:

$$(A - 6I_3)X = 0.$$

Consider the Augmented matrix

$$\left[\begin{array}{ccc|c} -1 & 0 & 1 & 0 \\ 0 & -8 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Eigenvector corresponding to eigenvalue $\lambda = 6$:

$$(A - 6I_3)X = 0.$$

Consider the Augmented matrix

$$\left[\begin{array}{ccc|c} -1 & 0 & 1 & 0 \\ 0 & -8 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

So, from here, we get $x_1 - x_3 = 0$ and $x_2 = 0$, i.e.,

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Eigenvector corresponding to eigenvalue $\lambda = 6$:

$$(A - 6I_3)X = 0.$$

Consider the Augmented matrix

$$\left[\begin{array}{ccc|c} -1 & 0 & 1 & 0 \\ 0 & -8 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

So, from here, we get $x_1 - x_3 = 0$ and $x_2 = 0$, i.e.,

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

An eigenvector corresponding to $\lambda = 6$ is $v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

Exercise 28:

Find all the eigenvalues of the matrix

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

Ans: $\pm 1, \pm i$.

Exercise 29:

Determine all 2×2 matrices A such that A has eigenvalues 2 and -1 with corresponding eigenvectors

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

respectively.

Ans: $\begin{bmatrix} 2 & -6 \\ 0 & -1 \end{bmatrix}$.

Exercise 30:

Find eigenvalues and eigenvectors of the following matrix.

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Ans: Eigenvalues $= 1, 3$, Eigenvector corresponding to eigenvalue $\lambda = 1$ is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and corresponding to $\lambda = 3$ is $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

Exercise 31:

Find the eigenvalues and eigenvectors of the following matrix.

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Ans: Eigenvalues $= i, -i$, Eigenvector corresponding to eigenvalue $\lambda = i$ is $\begin{bmatrix} i \\ 0 \end{bmatrix}$ and corresponding to $\lambda = -i$ is $\begin{bmatrix} 0 \\ -i \end{bmatrix}$.

Exercise 32:

For each of the following matrix, compute the characteristic polynomial, eigenvalue, and eigenvectors corresponding to each eigenvalue.

(a) $A = \begin{bmatrix} 3 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ (b) $B = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -2 & 2 \\ 3 & 0 & 1 \end{bmatrix}$

Ans:

(a) $p_A(\lambda) = (\lambda - 3)(\lambda + 1)(\lambda - 2)^2$, Eigenvalues $\lambda_1 = \lambda_2 = 2$, $\lambda_3 = -1$ and $\lambda_4 = 3$. Eigenvector $v_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $v_{-1} = \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \\ 1 \end{bmatrix}$

(b) $p_B(\lambda) = (\lambda - 4)(\lambda + 2)^2$, Eigenvalues $\lambda_1 = \lambda_2 = -2$ and $\lambda_3 = 4$. Eigenvector corresponding to $\lambda = -2$ is $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and corresponding to $\lambda = 4$ is $\begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$

Exercise 33:

Consider the 2×2 matrix

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix},$$

where θ is a real number $0 \leq \theta < 2\pi$.

(a) Find the characteristic polynomial of the matrix A.

(b) Find the eigenvalues of the matrix A.

(c) Determine the eigenvectors corresponding to each of the eigenvalues of A.

Ans: The characteristic polynomial of A is $p_A(\lambda) = \lambda^2 - (2\cos\theta)\lambda + 1$, eigenvalues $\cos\theta \pm i\sin\theta$.

Exercise 34:

Find all the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 3 & 9 & 9 & 9 \\ 9 & 3 & 9 & 9 \\ 9 & 9 & 3 & 9 \\ 9 & 9 & 9 & 3 \end{bmatrix}.$$

Ans:Eigenvalues are $30, -6$, eigenvector corresponding to $\lambda = 30$ is $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ and

corresponding to -6 $\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$.

Exercise 35:
Let

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Find the characteristic polynomial and all the eigenvalues (real and complex) of A.
Ans: The characteristics polynomial $-\lambda^3 + 1$ and eigenvalues are 1, $\frac{-1 \pm \sqrt{3}i}{2}$.

Trace

Definition (Trace)

Let A be a square matrix of size $n \times n$ then trace(A) is defined as the sum of diagonal elements of A.

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Example

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Then

$$\text{trace}(A) = 1 + 5 + 9 = 15$$

Properties of Eigenvalues and Eigenvectors

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- Let A be a square matrix of size $n \times n$ such that sum of each row (or each column) is same and is α. Then α is an eigenvalue of A.

- Eigenvectors corresponding to distinct eigenvalues are linearly independent.

- If $\lambda_1, \lambda_2 \ldots \lambda_n$ are the Eigenvalues of a $n \times n$ matrix A, then ${\lambda_1}^m, {\lambda_2}^m \ldots {\lambda_n}^m$ are the Eigenvalues of A^m **(m being a positive integer)**. Corresponding eigenvectors will be unchanged.

Cont. . .

- Let A be an invertible matrix and λ is an eigenvalue of A then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1}. Corresponding eigenvectors will be unchanged.

Cont...

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- If λ is an eigenvalue of A and $p(A)$ is a polynomial in A then $p(\lambda)$ is an eigenvalue of $p(A)$.

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- Eigenvalues of Hermitian matrices and symmetric matrices are real numbers.

- Eigenvalues of skew-Hermitian and skew symmetric matrices are purely imaginary and zero.

- Eigenvalues of unitary matrices and orthogonal matrices are of unit modulus.

- If A is a non invertible matrix, then **0** is an eigenvalue of A, i.e., if $\det(A) =$ then 0 is an eigenvalue of A.

Example

If $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{bmatrix}$. Find the eigenvalues of $A^3 + 5A + 8I$.

Solution

First, we find the eigenvalues of A. The characteristics polynomial of A is given by

$$p_A(\lambda) = \begin{vmatrix} -1-\lambda & 2 & 3 \\ 0 & 3-\lambda & 5 \\ 0 & 0 & -2-\lambda \end{vmatrix} = \lambda^3 - 7\lambda - 6 = (\lambda+1)(\lambda-3)(\lambda+2).$$

Thus, the roots of the characteristics polynomial are $-1, 3, -2$ and hence the eigenvalues of A are $-1, 3, -2$.

If λ is the eigenvalue of A, then by the property eigenvalues of $A^3 + 5A + 8I$ will be $\lambda^3 + 5\lambda + 8$. By putting $\lambda = -1, 3, -2$, Eigenvalues of $A^3 + 5A + 8I$ will be $2, 50, -10$.

Example

If the characteristic polynomial of a 3×3 matrix is $\lambda^3 - 4\lambda^2 + a\lambda + 30$, $a \in \mathbb{R}$, and one eigenvalue of A is 2, then find the other eigenvalues of A.

Solution

Given that $\lambda = 2$ is an eigen value. So, it must satisfy characteristic polynomial.

$$2^3 - 4 * 2^2 + 2a + 30 = 0 \Rightarrow a = -11$$

Characteristic polynomial :

$$\begin{aligned} \lambda^3 - 4\lambda^2 - 11\lambda + 30 \\ \implies (\lambda - 2)(\lambda - 5)(\lambda + 3) \\ \lambda_1 = 2, \lambda_2 = 5 \text{ and } \lambda_3 = -3 \end{aligned}$$

Exercise 36:

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be an 2×2 matrix.

Express the eigenvalues of A in terms of the trace and the determinant of A.

Ans: $\frac{\text{trace}(A)+\sqrt{\text{trace}(A)^2-4\det(A)}}{2}$, $\frac{\text{trace}(A)-\sqrt{\text{trace}(A)^2-4\det(A)}}{2}$.

Exercise 37:

Find the eigenvalues and eigenvectors of the following matrix.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Ans: Eigenvalues are $0, 3$ and eigenvectors corresponding to $\lambda = 0$ are $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and corresponding to $\lambda = 3$ is $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

Exercise 38:

Find the eigenvalues and eigenvectors of the following matrix.

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

Ans: Eigenvalues are $-1, 5$ and eigenvectors corresponding to $\lambda = -1$ are $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$ and corresponding to $\lambda = 5$ is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Exercise 39:

Let $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$. Find the eigenvalues of the matrix A. Also give the algebraic multiplicity of each eigenvalue.

Ans: The eigenvalues of A are 0 and 2 with algebraic multiplicities 3 and 1, respectively.

Exercise 40:

Let A be a 3×3 matrix. Suppose that A has eigenvalues 2 and -1, and suppose that $\mathbf{u}$ and $\mathbf{v}$ are eigenvectors corresponding to 2 and -1, respectively, where $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$. Then compute $A^5\mathbf{w}$, where $\mathbf{w} = \begin{bmatrix} 7 \\ 2 \\ -3 \end{bmatrix}$.

Ans: $\begin{bmatrix} 92 \\ -2 \\ -96 \end{bmatrix}$

Exercise 41:

Find all the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 3 & -2 \\ 6 & -4 \end{bmatrix}.$$

Ans: Eigenvalues are $0, -1$. Eigenvector corresponding to $\lambda = 0$ is $\begin{bmatrix} 2/3 \\ 1 \end{bmatrix}$ and corresponding to $\lambda = -1$ is $\begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$.

Exercise 42:

Let

$$A = \begin{bmatrix} 3 & -12 & 4 \\ -1 & 0 & -2 \\ -1 & 5 & -1 \end{bmatrix}.$$

Then find all eigenvalues of A^5. If A is invertible, then find all the eigenvalues of A^{-1}.

Ans: Eigenvalues of A^5 are $32, -1, 1$. Eigenvalues of A^{-1} are $\pm 1, \frac{1}{2}$.

Exercise 43:

Let

$$A = \begin{bmatrix} a & -1 \\ 1 & 4 \end{bmatrix}$$

be a 2×2 matrix, where a is some real number. Suppose that the matrix A has an eigenvalue 3.

(a) Determine the value of a.

(b) Does the matrix A have eigenvalues other than 3?

Ans: (a) $a = 2$

(b) No

Exercise 44:
Determine all eigenvalues and their algebraic multiplicities of the matrix

$$A = \begin{bmatrix} 1 & a & 1 \\ a & 1 & a \\ 1 & a & 1 \end{bmatrix},$$

where a is a real number.

Exercise 45:
Prove that the matrix

$$A = \begin{bmatrix} 1 & 1.00001 & 1 \\ 1.00001 & 1 & 1.00001 \\ 1 & 1.00001 & 1 \end{bmatrix}$$

has one positive eigenvalue and one negative eigenvalue.

Exercise 46:
Let A be a square matrix and its characteristic polynomial is given by $p(\lambda) = (\lambda - 1)^3(\lambda - 2)^2(\lambda - 3)^4(\lambda - 4)$. Find the rank of A.

Exercise 47:

Let n be an odd integer and let A be an $n \times n$ real matrix. Prove that the matrix A has at least one real eigenvalue.

Exercise 48:

Let λ be an eigenvalue of $n \times n$ matrices A and B corresponding to the same eigenvector x.

(a) Show that 2λ is an eigenvalue of $A + B$ corresponding to x.

(b) Show that λ^2 is an eigenvalue of AB corresponding to x.

Exercise 49:

Find the eigen values of $A^3 - 5A^2 + 7I$ if $A = \begin{bmatrix} 5 & -10 & -5 \\ 2 & 14 & 2 \\ -4 & -8 & 6 \end{bmatrix}$.

Exercise 50:

For the matrix $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$, find the eigenvalues of $3A^3 + 5A^2 - 6A + 2I$.